Algebraic Routing, Symbolic Cofactor Reduction, and Multiscale Coordinate Transduction in SHA-256 Inversion

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Paradigmatic Shift: From Candidate Enumeration to Dependency-Injected Routing

Cryptographic preimage recovery for the Secure Hash Algorithm (SHA-256) is conventionally formulated as an unstructured search across an exponentially large candidate space, requiring $O(2^n)$ forward evaluations to locate an input satisfying a target digest. When reformulated through the lens of algebraic topology and formal verification, however, hash inversion transforms from a brute-force guessing procedure into a deterministic routing and constraint-binding problem across coupled coordinate systems. Within this structural paradigm, a computational gap is not an empty volume to be searched by generating and testing candidate bitstrings; rather, it is a formally specified boundary—a *hole*—whose algebraic geometry is rigidly determined by the surrounding forward and backward transformations.

The operational semantics of this routing architecture rest on five foundational primitives that govern how information is written, stored, and recovered across computational scales. A *WHAT* is defined as a state write that intrinsically carries its own algebraic inverse, enforcing the invariant that no valid state transformation exists without a corresponding return path ("a way home"). A *WHERE* represents that identical underlying structure read at a spatial, angular, or temporal offset by a designated reader; within this framework, write and read operations are strictly offset and never concurrent. When the offset between the write and the reader is known, traversal constitutes a *ROUTE*, which resolves the target state in $O(n)$ steps proportional to the bit-length of the address. Conversely, when the offset is discarded or treated as unknown, recovery degenerates into a *SEARCH*, incurring a computational cost proportional to the

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cardinality of the state space $O(2^n)$. The fundamental unit of analytical work is the *HOLE*: a *WHAT* that has been committed to the system before its corresponding *WHERE* has been resolved by a reader.

Operational Primitive / Ledger State	**Structural Definition**	**Computational / Verification Semantics**
WHAT	A reversible write operation carrying its own algebraic inverse	Establishes a state transition with a guaranteed return route ("way home")
WHERE	The written structure interrogated at a non-concurrent offset by a reader	Separates the underlying binary substrate from the Equivalence Vocabulary of the probe
ROUTE	Directed traversal across a known coordinate offset	Resolves target locations in $O(n)$ steps proportional to address width
SEARCH	Unstructured enumeration across an unknown coordinate offset	Scales with domain cardinality $O(2^n)$; represents the failure mode of unindexed reading
HOLE	A sealed *WHAT* written prior to the resolution of its *WHERE*	Defines the exact boundary shape that the dependency-injection chain must bind
BOUND	Single algebraic provider uniquely determines the variable	Dependency-injection container resolves the variable without branching
AMBIGUOUS	Multiple providers satisfy the local constraint	Requires an independent orthogonal readout as a qualifying selector
UNBOUND	No active provider reaches the variable under current elimination rules	Signals that the container is naming the next algebraic link to be constructed

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Operational Primitive / Ledger State	Structural Definition	Computational / Verification Semantics
STANDS / RUNS	External theorem/measurement (*STANDS*) or pre-sealed predictive execution (*RUNS*)	Replaces heuristic confirmation with cryptographically sealed replay and landing checks
DEAD END / OPEN	Route lands on an external owner (*DEAD END*) or awaits a missing chain link (*OPEN*)	Preserves all computed trajectories by re-routing misattributed landings to their true providers

To prevent premature collapse of the state space, constraint resolution operates as a strict Dependency-Injection (DI) container. Rather than guessing values to fill a hole, the procedure subtracts every dependency already accounted for by the forward construction and inspects the algebraic remainder. In coupled discrete systems, this remainder is never an arbitrary residual; it is an earlier structural layer re-addressed and read by an opposing wave. This principle is directly analogous to shell closure in nuclear and atomic physics. In the nuclear shell model, the gap between magic number 14 and the calcium floor at 20 is governed by relativistic pseudospin symmetry in the lower Dirac component. As oscillator shell $n+1$ unfolds, its highest-angular-momentum orbital (top-j) lacks a pseudospin partner because such a partner would require radial quantum number $n_r = 0$. This "homeless" top-j orbital drops down to close the lower shell, leaving a remainder in shell $n+1$ whose state count exactly equals the full count of shell n. Inverting the filling rule by reading the anti-aligned $2s_{1/2}$ doublet first yields $14+2=16$, followed by $+4=20$, causing the subshell closure at 16 to emerge algebraically without candidate search. An identical structural mechanism drives the Madelung period-doubling sequence $(2,2,8,8,18,18,32,32)$ in atomic structure, where each period copies the preceding group one principal quantum number n higher, augmented by at most one nodeless orbital ($n = l+1$) that possesses no radial predecessor and arrives every second group.

Applying this subtraction discipline to SHA-256 isolates the exact locus of cryptographic hardness. The forward construction and its known local inverses already account for Davies–Meyer feed-forward subtraction of the initial chaining vector (H_0), deterministic 64-round state inversion when the message schedule W and round constants K are given, and exact bijective recovery of the entire 64-word message schedule from its final 16 words. When these accounted-for components are subtracted, the remainder is identified precisely: it is the nonlinear Boolean elimination rule required to route carry and borrow dependencies across the schedule-round interface, combined with the fiber selector that distinguishes individual preimages within a shared digest fiber.

Symbolic Inverse Compilation and ROBDD Cofactor Collapse

To execute the inverse route without enumerating candidate messages or expanding full-domain bitmasks, the 64 reverse rounds of SHA-256 and the backward message schedule are compiled into a shared symbolic Directed Acyclic Graph (DAG) over the Boolean basis $\{\oplus, \wedge, 1\}$. Within this compiler architecture (shared_carry_inverse.py, sealed prior to digest linking under SHA-256 hash 6b9a7d6549cd336c97a884fe85f087e6d67c542254a4c5f443a54df41cccd8b5), every working-state bit and schedule bit is instantiated as a pointer to a canonical gate node, where structural hashing guarantees that

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identical Boolean sub-expressions reuse the exact same node in memory. Bitwise rotations (ROTR, Σ_0, Σ_1, σ_0, σ_1) reduce to zero-cost pointer rewiring, while the initial hash values H_0, round constants $K_0 \ldots K_{63}$, and SHA-256 padding bits are injected as fixed terminal nodes.

Modular addition ($x + y \bmod 2^{32}$) and modular subtraction ($x - y \bmod 2^{32}$) are embedded inside the symbolic graph through their exact carry and borrow factorizations. At each bit position $j \in \{0, \ldots, 31\}$, addition defines the local propagate and generate factors:

$$p_j = x_j \oplus y_j, \qquad g_j = x_j y_j$$

from which the sum bit and outgoing carry bit are constructed recursively:

$$s_j = p_j \oplus c_j, \qquad c_{j+1} = g_j \oplus p_j c_j$$

Unrolling this recurrence yields the explicit path-expanded carry chain across all lower bit positions:

$$c_{j+1} = \oplus_{i=0}^{j} g_i \prod_{k=i+1}^{j} p_k \ \oplus \ c_0 \prod_{k=0}^{j} p_k$$

Every monomial in this expansion represents a discrete carry-generation event g_i that survives an unbroken cascade of subsequent propagation gates p_k. Subtraction retains the exact dual borrow architecture, substituting the local borrow-generate factor $h_j = (1 \oplus x_j) y_j$ and borrow-propagate factor $q_j = 1 \oplus x_j \oplus y_j$. When lifted from $\mathbb{F}_2$ to the integers $\mathbb{Z}$ via high-precision PSLQ integer-relation recovery (certified at 100 to 300 decimal digits of precision) and full 8-row truth-table Möbius inversion over the local basis $\mathcal{B} = (1, x, y, xy, c, xc, yc, xyc)$, these single-bit transitions correspond to exact pointwise polynomial identities:

$$\text{carry}(x, y, c) = xy \oplus xc \oplus yc = xy + xc + yc - 2xyc$$

$$\text{borrow}(x, y, c) = (1 \oplus x)y \oplus (1 \oplus x \oplus y)c = y + c - xy - xc - yc + 2xyc$$

Symbolic add and subtract modules built from these factorizations were verified across 1,000 random 32-bit operand pairs, matching standard modular arithmetic with zero discrepancies. Composing all 64 inverse rounds from the digest-derived terminal state back to round 0 imposes a system of 256 Boolean closure equations on the unknown message-bit vector μ:

$$\Psi_H(\mu) = 0$$

To evaluate the reach of purely local constraint propagation, the compiler was tested against two held digests generated from the 1-byte ASCII source Q (0x51, 8 unknown message bits) and the 2-byte ASCII source GO (16 unknown message bits), withholding the source strings from the compiler until the post-compilation replay audit.

The execution metrics from the symbolic DAG compiler reveal a critical structural phenomenon. Doubling the unknown message interface from 8 bits (Q) to 16 bits (GO) increases the total number of unique XOR/AND nodes by less than 0.9%—from 117,240 to 118,294—while performing 34,491 and 33,509 constant and identity simplifications in roughly 0.14 seconds. When evaluated at the true source messages during the post-run replay check, both graphs collapse cleanly to 256 zero residual bits, confirming that the symbolic inverse captures the exact round semantics, schedule dependencies, and carry/borrow chains without loss. However, when subjected to local forced-consequence propagation—where an XOR gate resolves its third terminal only if two terminals are known, and an AND gate forces both inputs to 1 if its

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output is 1 or forces output 0 if either input is 0—the propagator assigns 2,717 internal nodes in the 1-byte graph and 2,247 internal nodes in the 2-byte graph, yet binds **zero** message bits.

Symbolic DAG & ROBDD Execution Metric	**1-Byte Interface (Q, 0x51)**	**2-Byte Interface (GO)**	**Structural Interpretation**
Unknown Message Bits (μ)	8	16	Input degrees of freedom in W_0 [cite: 1, 6]
Unique XOR/AND DAG Nodes	117,240	118,294	Sub-linear node growth (+0.90%) due to structural gate sharing
Reused Gate Requests	3,661	3,589	Identical sub-expressions merged during DAG construction
Constant / Identity Simplifications	34,491	33,509	Fixed padding and round constants folded at compile time
DAG Compile Time	0.143 s	0.140 s	Direct symbolic graph construction without candidate search
Forced Local Propagation Time	0.131 s	0.119 s	Local unit-clause resolution + replay verification
Internal Nodes Bound (Local Propagator)	2,717	2,247	Boundary constants penetrate initial reverse layers locally
Message Bits Bound (Local Propagator)	0	0	Local rules cannot cross reconvergent nonlinear carry/Ch/Maj loops

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Symbolic DAG & ROBDD Execution Metric	1-Byte Interface (Q, 0x51)	2-Byte Interface (GO)	Structural Interpretation
ROBDD Cofactor Collapse (Peak / Total Nodes)	80,556–80,788 / ~ 2.73×10^6	*Open scaling target*	Canonical cofactor elimination resolves all reconvergent dependencies
Message Bits Bound (ROBDD Conjunction)	8/8 (0x51 at constraint 7)	*Target for region pruning*	Exact global Boolean intersection isolates unique preimage

The failure of local propagation to bind μ does not indicate ambiguity in the equation $\Psi_H(\mu) = 0$; rather, it reflects the topological limitation of tree-local inference in a graph with massive reconvergent fanout. In SHA-256, each unknown bit μ_j branches rapidly through the message schedule (σ_0, σ_1) and nonlinear round functions ($\mathrm{Ch}, \mathrm{Maj}$, and modular carry chains) before reconverging at the 256 endpoint equations. Local propagation treats distinct paths sharing a common latent variable as independent until a single-gate forcing condition is met, stalling whenever correlation information is distributed across sums of products.

Eliminating this stalemate requires an exact elimination rule that preserves global Boolean location conditions across shared nonlinear factors. This missing link is realized by compiling the symbolic DAG into a Reduced Ordered Binary Decision Diagram (ROBDD) via Shannon cofactor expansion (cofactor_inverse_bdd.py). Each ROBDD node encodes a decision triple $(v, \mathrm{low}, \mathrm{high})$ over a fixed variable ordering of μ, enforcing two canonical reduction invariants: if $\mathrm{low} = \mathrm{high}$, the redundant decision node is eliminated, and if an identical triple $(v, \mathrm{low}, \mathrm{high})$ already exists in the unique table, its node pointer is reused. When two Boolean functions F and G are combined under a binary operation (such as $\oplus$ during round construction or $\wedge$ when intersecting endpoint return constraints), the engine recursively decomposes them along the top variable μ_j:

$$F = (1 - \mu_j)F_0 + \mu_j F_1, \qquad G = (1 - \mu_j)G_0 + \mu_j G_1$$

$$F \wedge G = (1 - \mu_j)(F_0 \wedge G_0) + \mu_j (F_1 \wedge G_1)$$

Unlike local gate propagation, recursive cofactor reduction forces identical variables along every reconvergent carry and schedule path to take the exact same truth value simultaneously within each branch $F_0 \wedge G_0$ and $F_1 \wedge G_1$, immediately collapsing contradictory sub-graphs to the terminal node 0.

Return Constraint Step	Target Word 0 Bit Condition (0x6A09E667)	Surviving Assignments	Active ROBDD Decision Nodes	Cumulative Pruning Ratio
Constraint 0	None (unconstrained 8-bit interface)	256	0	$1.00 \times$

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Return Constraint Step	Target Word 0 Bit Condition (0x6A09E667)	Surviving Assignments	Active ROBDD Decision Nodes	Cumulative Pruning Ratio
				[cite: 1, 7]
Constraint 1	Word 0, Bit $0 = 1$	131	76	$1.95 \times$ [cite: 1, 7]
Constraint 2	Word 0, Bit $1 = 1$	56	63	$4.57 \times$ [cite: 1, 7]
Constraint 3	Word 0, Bit $2 = 1$	28	46	$9.14 \times$ [cite: 1, 7]
Constraint 4	Word 0, Bit $3 = 0$	12	38	$21.33 \times$ [cite: 1, 7]
Constraint 5	Word 0, Bit $4 = 0$	6	25	$42.67 \times$ [cite: 1, 7]
Constraint 6	Word 0, Bit $5 = 1$	3	16	$85.33 \times$ [cite: 1, 7]
Constraint 7	Word 0, Bit $6 = 1$	1	8	$256.00 \times$ (Unique closure)
Constraints 8–256	Word 0, Bit 7 through Word 7, Bit 31	1	8	$256.00 \times$ (Verified consistent)

When all 64 inverse rounds are compiled for the 1-byte target digest and intersected sequentially with the required initial chaining state H_0 (beginning with Word 0 = 0x6A09E667, least-significant bit first), the candidate space collapses exponentially ($256 \rightarrow 131 \rightarrow 56 \rightarrow 28 \rightarrow 12 \rightarrow 6 \rightarrow 3 \rightarrow 1$). By the seventh bit constraint of Word 0, every incompatible path has been annihilated, leaving a single satisfying assignment

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that persists unchanged across the remaining 249 return-bit checks. Tracing the unique surviving path from LSB (μ_0) to MSB (μ_7) shows that branch 0 has zero solutions and branch 1 has one solution at μ_0, μ_4, μ_6, whereas branch 0 has one solution and branch 1 has zero solutions at $\mu_1, \mu_2, \mu_3, \mu_5, \mu_7$. This yields the exact canonical Boolean characteristic polynomial:

$$\Phi_H(\mu) = \mu_0(1 \oplus \mu_1)(1 \oplus \mu_2)(1 \oplus \mu_3)\mu_4(1 \oplus \mu_5)\mu_6(1 \oplus \mu_7)$$

Reading the forced bits in standard MSB-first byte order ($\mu_7\mu_6\mu_5\mu_4\mu_3\mu_2\mu_1\mu_0$) recovers the binary word 01010001 = 0x51 = 'Q'. Independent compilations across four distinct variable orderings—LSB-first (0,1,2,3,4,5,6,7), MSB-first (7,6,5,4,3,2,1,0), Outside-inward (0,7,1,6,2,5,3,4), and Even-then-odd (0,2,4,6,1,3,5,7)—demonstrate that peak intermediate ROBDD node counts remain invariant within 0.29% (80,556 to 80,788 nodes, creating ~ 2.72–2.73×10^6 total nodes in 8.25–8.49 s). This invariance occurs because full 64-round SHA-256 diffusion mixes all 8 input bits to maximal algebraic degree, meaning every variable ordering encounters the same intermediate Boolean complexity before the endpoint constraints collapse the graph to 8 nodes.

The Nested Binary Ring: Spatial Reflection, Partner Involution, and Multiscale Ancestry

The message bits bound by the inverse graph are not isolated integers; they inhabit a nested, multiscale geometric coordinate system defined by the binary ring R_L. At each width $L \geq 1$, the ring arranges all $N = 2^L$ binary configurations into a circular sequence of $M = 2^{L-1}$ distinct partner pairs such that every configuration occupies a unique angular address with zero duplicates.

The algebraic engine of the ring is the conditional partner involution $P_L: \{0,1\}^L \to \{0,1\}^L$, which pairs each word with its bit-reversal unless the word is a palindrome, in which case it pairs the word with its bitwise one's complement:

$$P_L(x) = \begin{cases} \text{reverse}(x), & \text{reverse}(x) \neq x, \\ \text{complement}(x) = x \oplus 1^L, & \text{reverse}(x) = x. \end{cases}$$

Because bit-reversal is an involution on non-palindromes and bitwise complementation maps palindromes strictly to distinct palindromes of the same width, P_L is a fixed-point-free involution across all positive widths:

$$P_L(P_L(x)) = x, \qquad P_L(x) \neq x \quad \forall x \in \{0,1\}^L$$

The ring ordering grow(L) constructs the sequence by sorting the 2^{L-1} canonical pair representatives, inserting each representative and its partner from the center seam outward, and applying a final center swap across the seam to preserve orientation. For width $L = 3$, this generates the exact eight-word cycle 000 001 010 110 | 011 101 100 111, where the center seam separates 110 and 011.

Direct computational verification across all ring widths from $L = 1$ through $L = 12$ establishes that the second half of every ring is not an independent sequence; it is the exact spatial reflection of the first half transformed simultaneously by the partner involution, including the two center positions affected by the seam swap. Defining the spatial address reflection on $\{0, \ldots, N-1\}$ by $\sigma_L(i) = N - 1 - i$, the ring satisfies the universal intertwining identity:

$$R_L(\sigma_L(i)) = P_L(R_L(i)) \quad \Leftrightarrow \quad R_L \circ \sigma_L = P_L \circ R_L$$

Because both $\sigma_L^2 = I$ and $P_L^2 = I$, the ring ordering acts as an exact equivariant bridge translating spatial reflection in index space into the conditional partner involution in word space. A naive geometric fold would superimpose identical words and leave the complementary half unpopulated; the two-step rule—reflecting

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the address $i \mapsto N - 1 - i$ while applying P_L—populates the complete state space from a single canonical half:

One Ordered Half + Address Reflection (σ_L) + Partner Map (P_L) + Center Swap = Complete Ring R_L

For the 1-byte SHA-256 preimage recovered by the ROBDD engine (0x51 = 'Q'), executing the original grow(8) generator places 01010001 at exact ring address $i = 70$ ($R_8[70] = 01010001$, flanked by 0x4E at position 68, 0x4F at 69, 0x53 at 71, and 0x55 at 72), corresponding to flattened bit slice $8 \cdot 70 = 560$ and angular sector $u = \frac{70}{256} = \frac{35}{128}$. Its reflected address $\sigma_8(70) = 255 - 70 = 185$ holds the exact reversed partner $R_8[185] = 10001010 = P_8(01010001)$. Furthermore, explicit rank and unrank algorithms now compute the position of any word or reconstruct the word at any index without materializing the array, passing exhaustive checks through width 8 and 512-bit round-trip partner checks. Thus, at cryptographically relevant widths ($L = 512$ or $L = 2048$), the ring is never held in memory; it is only addressed.

Pair Residuals, Flattened Mirror Invariants, and Ancestry Generating Functions

Comparing a ring word with its reflected partner via bitwise XOR defines the pair residual operator:

$$Q_L(x) = x \oplus P_L(x)$$

This residual satisfies two structural invariants: it is side-invariant ($Q_L(P_L(x)) = Q_L(x)$) and it is unconditionally a palindrome ($Q_L(x) = \mathrm{reverse}(Q_L(x))$), evaluating to $x \oplus \mathrm{reverse}(x)$ for non-palindromes and to the all-ones word 1^L for palindromes. Consequently, when the entire ring R_L is flattened into a continuous bitstring S_L of length $L \cdot 2^L$, reflecting across the flattened bit index $q^\star = L \cdot 2^L - 1 - q$ yields the exact cross-width mirror law:

$$S_L[q] \oplus S_L[q^\star] = \begin{cases} 1, & \text{if the word containing } q \text{ is a palindrome,} \\ 0, & \text{otherwise.} \end{cases}$$

The mirror-XOR of the flattened ring acts as an exact palindrome detector mask, marking one full L-bit block of ones for every palindromic word and zeros elsewhere. However, the residual alone is a many-to-one projection. Exact fiber counting across widths $L = 1 \ldots 12$ proves that the number of distinct residuals is $2^{\lceil L/2 \rceil}$ when L is odd (where every residual fiber contains exactly $2^{\lceil L/2 \rceil}$ preimages) and $2^{L/2} - 1$ when L is even (where the all-ones palindrome residual 1^L absorbs twice the standard fiber cardinality, depressing the Shannon entropy of the residual-only readout strictly below $L/2$ bits). Retaining full invertibility requires pairing the residual with the ring address or side selector.

Ring Structure / Coordinate System	**Exact Algebraic Formula / Generating Function (|z|<1/2\vert{}z\vert{} < 1/2|z|<1/2)**	**Operational Role & Verification Scope**
Total Words $V(z)$	$V(z) = \frac{2z}{1-2z}, \quad [z^L]V(z) = 2^L$	Complete binary state space at width L [cite: 1, 4]
Partner Pairs $A(z)$	$A(z) = \frac{z}{1-2z}, \quad [z^L]A(z) = 2^{L-1}$	Canonical half-ring cardinality under fixed-point-free P_L

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Ring Structure / Coordinate System	Exact Algebraic Formula / Generating Function (|z|<1/2\vert{}z\vert{} < 1/2|z|<1/2)	Operational Role & Verification Scope
		[cite: 1, 4]
Palindromes $C(z)$	$C(z) = \frac{2z(1+z)}{1-2z^2}, \quad [z^L]C(z) = 2^{\lceil L/2 \rceil}$	Words taking the complement branch $P_L(x) = x \oplus 1^L$ [cite: 1, 4]
Flattened Bit Slices $S(z)$	$S(z) = \frac{2z}{(1-2z)^2}, \quad [z^L]S(z) = L \cdot 2^L$	Total angular slice resolution of flattened string S_L [cite: 1, 4]
New Ancestry Roots $G(z)$	$G(z) = \frac{z(1-z^2)}{1-2z}$	Irreducible canonical roots not entered from width $L-2$ [cite: 1, 4]
Bivariate Ancestry Forest $A(z,t)$	$A(z,t) = \frac{G(z)}{1-tz^2} = \frac{z(1-z^2)}{(1-2z)(1-tz^2)}$	Tracks continuation depth k via t^k; recovers $A(z)$ at $t=1$ [cite: 1, 4]
Ancestry Continuation Map $F_L(a)$	$F_L(a) = \min(1a1, P_{L+2}(1a1)), \quad x = P^{\varepsilon}(F^k(\rho))$	Bijective root–depth–side representation ($\|x\| = \|\rho\| + 2k$)

Beyond single-ring addressing, words are linked across scales by the padding continuation operator $F_L(a) = \min(1a1, P_{L+2}(1a1))$, which embeds a canonical representative from width L into width $L+2$. Every nonempty binary word decomposes uniquely into root–depth–side coordinates (ρ, k, ε) such that $x = P^{\varepsilon}(F^k(\rho))$ and $|x| = |\rho| + 2k$, verified exhaustively across all $8{,}190$ nonempty words for $L = 1 \ldots 12$ and on 512-bit continuations from width-2 roots.

The Common-Angle Radial Reader and Continuation Grammar

While ancestry coordinates track discrete steps of $L \mapsto L+2$, the radial reader interrogates the entire concentric stack simultaneously along a single continuous angular coordinate $u \in [0,1)$. At ring L, the reader selects word index $i_L = \lfloor 2^L u \rfloor$ and internal bit slice $k_L = \lfloor L\{2^L u\} \rfloor$, extracting the radial bit:

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$$b_L(u) = R_L[\lfloor 2^L u \rfloor][\lfloor L\{2^L u\} \rfloor]$$

Each finite radial prefix $p \in \{0,1\}^k$ corresponds to a support set $A_p \subseteq [0,1)$ of compatible angular intervals; appending a bit at depth $k+1$ intersects the parent support set with the slice boundaries of the next ring ($A_{p0} = A_p \cap B_{k+1,0}$ and $A_{p1} = A_p \cap B_{k+1,1}$). Grouping prefixes by their permitted future suffixes $\mathcal{C}_D(p) = \{z : |pz| \le D, A_{pz} \neq \varnothing\}$ via the Myhill–Nerode equivalence relation $p \equiv_D q \Leftrightarrow \mathcal{C}_D(p) = \mathcal{C}_D(q)$ compresses the entire depth-10 radial continuation tree into exactly **75 viable continuation states** plus 1 rejection state. However, exact interval propagation proves that finite-horizon exclusions do not saturate: the 32 minimal forbidden prefixes identified through depth 10 are followed by 21 new minimal exclusions at depth 11 and 48 more at depth 12. Because two prefix sets of identical measure can intersect subsequent ring boundaries differently, the exact reader state must retain the geometric interval union $I(p)$ rather than collapsing to scalar angle counts.

This interval geometry closes the pre-p segment of the benchmark radial target profile $q_D = 100\,1^{D-5}00$. For depths $D \equiv 0,1 \pmod 3$, setting $p = \lfloor 2D/3 \rfloor + 1$ and the certificate angle $u_D = \frac{1}{2} + 2^{-p} + \frac{2}{3}2^{-(D-1)}$ yields $2^L u_D = 2^{L-1} + \delta_L$ with $\delta_L = 2^{L-p} + \frac{2}{3}2^{L-D+1} \in (0,1)$ for all $L < p$. This locks the word address to the center word immediately clockwise of the seam, $i_L = 2^{L-1}$, and selects bit slice $k_L = \lfloor L\delta_L \rfloor$. For $L = 1,2,3$, $k_L = 0$ on the center words $C_1 = 1$, $C_2 = 01$, and $C_3 = 011$, producing the initial prefix 100. For $L \ge 4$, the center word obeys the exact closed form $C_L = 1^{\lfloor L/2 \rfloor - 1}01^{\lceil L/2 \rceil}$, whose sole zero sits at index $a_L = \lfloor L/2 \rfloor - 1$. For $4 \le L \le p-2$, δ_L bounds $k_L < a_L$ (landing in the leading run of ones), while at $L = p-1$ (for $D \ge 13$), $k_{p-1} = \lfloor (p-1)/2 \rfloor = a_{p-1} + 1$, jumping immediately past the sole zero into the trailing run of ones. Thus the pre-p radial profile is algebraically derived, isolating the middle/terminal generated-word bit proofs ($p \le L \le D$) and the uniform $D \equiv 2 \pmod 3$ witness schedule as the remaining open radial links.

The Rolling DI Chain and the Boolean Möbius Transducer Bridge

The dynamic mechanism that transports values across scales and angular positions is exemplified by the rolling DI chain on the 8-bit circular source cycle $s = (0,0,0,1,1,0,1,1) = 0x1B$. In the reference audit workbook (two_bit_circle.xlsx), the clockwise decimal sequence (Column B) and counterclockwise decimal sequence (Column G) are not independent tables; because the reverse-read cycle 11011000 is an exact 3-position cyclic rotation of 00011011, the two columns obey the universal phase-shift identity across every window width L:

$$G_{L,i} = B_{L,(i+3) \bmod 8}$$

Furthermore, since the cycle contains exactly four zeros and four ones, an equally weighted aggregate readout across all eight phases yields $\frac{4}{8} = \frac{1}{2}(0.5)$, establishing that a "grey" or probabilistic value of $1/2$ emerges as a reader's collapsed average over an entirely deterministic binary substrate.

Letting $W_{L,i}$ denote the integer value of the L-bit cyclic window starting at angular index i, the stack evolves along two orthogonal axes that reuse the exact same stored prefix:

$$\text{Radial Growth (Width } L \to L+1\text{):} \qquad W_{L+1,i} = 2W_{L,i} + s_{(i+L) \bmod 8}$$

$$\text{Angular Sliding (Phase } i \to i+1\text{):} \qquad W_{L,i+1} = 2W_{L,i} - 2^L s_i + s_{(i+L) \bmod 8}$$

At full cycle width $L = 8$, the exiting bit and entering bit coincide ($s_{(i+8) \bmod 8} = s_i$), reducing angular rotation to exact multiplication by two modulo 255:

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$$W_{8,i+1} = 2W_{8,i} - 255s_i \equiv 2W_{8,i} \pmod{255}$$

All 56 rolling-window recurrence cells across widths and phases passed computational audit with zero errors. This audit also resolves the **5-bit nibble rollover phenomenon**. When two adjacent windows A_L and B_L are paired, their bit-exact concatenation is $C_L = 2^L A_L + B_L$. At $L = 4$, each window is a 4-bit hexadecimal nibble, so concatenating their printed hex labels matches C_4 identically. At $L = 5$, however, each window carries its own entering bit, expanding the pair by two bits per round into a 10-bit field. For $A_5 = 01101_2 =$ 0xD and $B_5 = 11011_2 =$ 0x1B, the bit-exact concatenation is $0110111011_2 =$ 0x1BB (443_{10}), whereas string concatenation of the hexadecimal labels produces D1B (0xD1B $= 3355_{10}$). Tracking bit-exact widths rather than variable-length printed labels keeps the rolling DI stack exact across all scales.

Explicit Operator Commutation Between Circular Motion and the Difference Engine

The structural bridge linking the rolling DI chain to the Boolean Möbius transform (Algebraic Normal Form / subset-XOR difference coordinates) is established by conjugating the circular tape operators over a power-of-two tape length $N = 2^m$. For a circular bit vector $x = (x_0, \dots, x_{N-1}) \in \mathbb{F}_2^N$, the Boolean Möbius transform $a = Mx$ is defined via the Pascal-mod-2 subset-lattice sum:

$$a_i = \bigoplus_{t \subseteq i} x_t = \bigoplus_{j=0}^{i} \left(\binom{i}{j} \bmod 2 \right) x_j$$

where $t \subseteq i$ denotes that every binary bit set in index t is also set in i. Because $M^2 = I$ over $\mathbb{F}_2$, the transform is strictly self-inverse ($x = Ma$).

Let R denote the single-step clockwise circular shift operator, $(Rx)_i = x_{(i+1) \bmod N}$. Conjugating R by M yields the exact transformed shift operator $T = MRM$. By Lucas's theorem and Pascal's identity modulo 2, for any index $i < N - 1$, summing adjacent Möbius coefficients telescopes the binomial coefficients:

$$a_i \oplus a_{i+1} = \bigoplus_j \left[\binom{i}{j} + \binom{i+1}{j} \right] x_j = \bigoplus_j \binom{i}{j-1} x_j = \bigoplus_{k=0}^{i} \binom{i}{k} x_{k+1} = (MRx)_i$$

At the top boundary index $i = N - 1$ (whose binary representation is 1^m), every index $j \in \{0, \dots, N-1\}$ satisfies $j \subseteq N - 1$, so $a_{N-1} = \bigoplus_{j=0}^{N-1} x_j$ is the global XOR parity of the entire circular tape, which is invariant under circular rotation. Consequently, circular transport R becomes a local nearest-neighbor XOR difference update with a held boundary value:

$$a'_i = (Ta)_i = \begin{cases} a_i \oplus a_{i+1}, & 0 \le i < N-1, \\ a_{N-1}, & i = N-1. \end{cases}$$

Writing S for the nilpotent upper-diagonal shift matrix ($S_{i,i+1} = 1$, $S^N = 0$), the operator takes the closed form $T = I + S$. Because $N = 2^m$, the binomial expansion over $\mathbb{F}_2$ satisfies $T^N = (I + S)^{2^m} = I + S^{2^m} = I$, proving that the 8-step circular return of the tape is algebraically carried by the nilpotent structure of the local difference stack. Moreover, retaining the boundary parity a_{N-1} makes the local difference update immediately invertible via suffix summation: $a_i = \bigoplus_{j=i}^{N-1} a'_j$.

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Circular Tape Operator (Bit Domain xxx)	Transformed Möbius Operator (ANF Domain a=Mxa = Mxa=Mx)	Commutation & Algebraic Identities	Empirical Verification Audit
Circular Shift R: $(Rx)_i = x_{(i+1) \bmod N}$	**Local Difference** $T = I + S$: $a'_i = a_i \oplus a_{i+1}$ $(i < N-1)$, $a'_{N-1} = a_{N-1}$	$MR = TM, \quad T^N = I, \quad a_i = \bigoplus_{j=i}^{N-1} a'_j$	256/256 bytes; 32,768/32,768 window reads ($L = 1 \ldots 16$)
Tape Reversal J: $(Jx)_i = x_{N-1-i}$	**Upper Subset-XOR** U: $(Ua)_i = \bigoplus_{j \supseteq i} a_j$	$MJ = UM, \quad U^2 = I, \quad UTU = T^{-1}$	100% pass across $N = 2^0 \ldots 2^{10}$ (100 trials/length)
Bitwise Complement: $x \mapsto x \oplus 1^N$	**Constant-Bit Toggle:** $a \mapsto a \oplus e_0$, where $e_0 = (1,0,\ldots,0)$	$M(x \oplus 1^N) = Mx \oplus e_0$	Exact across all power-of-two tape lengths
Conditional Partner $P(x)$	**Transformed Partner** $\tilde{P}(a)$: Ua if $Ua \neq a$, else $a \oplus e_0$	$MP = \tilde{P}M, \quad \tilde{P}^2 = I$	256/256 exact match on all 8-bit sources
Pair Residual $Q(x) = x \oplus P(x)$	**Transformed Residual:** $a \oplus \tilde{P}(a)$	$M(x \oplus P(x)) = a \oplus \tilde{P}(a)$	Exact preservation of palindrome residual class

Tape reversal J and the conditional partner involution P also possess exact closed-form conjugates in Möbius coordinates. Reversing the tape indices conjugates to the upper superset-XOR transform $U = MJM$, defined by $(Ua)_i = \bigoplus_{j \supseteq i} a_j$, which satisfies $U^2 = I$ and inverts the difference update via dihedral conjugation $UTU = T^{-1}$. Because complementing all N bits of the tape toggles only the constant ANF coefficient a_0 ($M(x \oplus 1^N) = Mx \oplus e_0$), the conditional partner involution transforms into $\tilde{P}(a) = Ua$ when $Ua \neq a$, and $\tilde{P}(a) = a \oplus e_0$ when $Ua = a$, satisfying $MP = \tilde{P}M$.

Homological Cut Algebra, Carry-Run Depth, and the Emergence of Directionality

To understand why local carry structures resist naive backward inversion while admitting compact forward summaries, binary addition must be examined through the homological algebra of its transition complex. When n-bit modular addition is modeled as a cubical complex generated by elementary bit increments $+2^i$ and $+2^j$ $(i < j)$, the local commutativity square at basepoint x is declared local (independent) if and only if the intermediate bits do not propagate a carry across the seam j:

$$D_{ij}(x) = 1 \Leftrightarrow \text{bits } i, \ldots, j-1 \text{ of } x \text{ are not all 1}$$

Because $D_{ij}(x)$ inspects only bit positions $0, \ldots, n-2$ below the top bit, the declaredness predicate factors through the projection $\mathbb{Z}/2^n \to \mathbb{Z}/2^{n-1}$. However, at the second homological level, a non-trivial structural

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obstruction emerges: the cycle space of undeclared 2-cells is not spanned by the boundaries of declared 3-cubes ($Z_2 \not\subseteq \mathrm{span}(\partial\mathcal{L}_3)$).

Through downward induction on the seam index j, the homology class $[r_{i,j}(x)]$ of any undeclared square reduces to a universal normal form governed by low-bit transport (Lemma L1) and seam-sign staircase decomposition (Lemma L2). Under Lemma L1, clearing the least-significant set bit below index i via the 3-cube boundary identity modifies the square solely by declared faces, proving that $[r_{i,j}(x)]$ is completely independent of all bits below i. Under Lemma L2, at the top bit ($j = n-1$), the increment is an involution ($A_{n-1}^{-1} = A_{n-1}$), yielding the exact base class $[r_{i,n-1}(x)] = (-1)^{x_{n-1}} c_{i,n-1}$. Below the top bit ($k < j \le n-2$), telescoping the transfer operator T_j along a staircase of bit clearings satisfies the width-independent algebraic identity:

$$\partial r_{i,j+1}(z) + T_j(z + 2^i) - T_j(z) = S_{i,j+1}(z) - S_{j,j+1}(w) - \sum_{k=i}^{j-1} \left[S_{kj}(u_k) + S_{kj}(u_k + 2^j) - S_{k,j+1}(u_k) \right]$$

Every term on the right-hand side resides strictly inside the declared disjoint-support span B_D (provided basepoints are anchored at the seam-zero convention where $x \oplus 2^j = x - 2^j$ when $x_j = 1$). Consequently, the homology class of an undeclared square depends on only two structural parameters—**seam parity** and **carry-run depth**—ignoring the exponentially large remainder of the basepoint label. Furthermore, the pair module M_{ij} generated by carry runs Q_{ij}^m ($m = 0, \ldots, n-j-1$) undergoes a unit-triangular basis change $Q_{ij}^m - Q_{ij}^{m-1} = c_{i,j+m} - c_{j,j+m}$ against the canonical generators c_{ab}, proving the exact dimension theorem:

$$\dim M_{ij} = n - j$$

Geometrically, the dimension of the (i, j) pair module equals the exact number of possible carry-run continuation depths above the seam j.

State Formation and Backward Non-Invertibility

When a reader interrogates an n-bit addition across a subset of read bit positions (a cut $S \subseteq \{0, \ldots, n-1\}$), the quotient summary $\Phi_S(\alpha)$ compresses the read history into contiguous carry-run intervals characterized by their transfer types (kill, set, or transparent). Crucially, Φ_S alone cannot update itself when a new bit position $p \notin S$ is read; the update operator $U(\Phi_S(\alpha), S, p, u)$ requires the cut geometry S to determine whether position p opens a new isolated run, extends an adjacent run, or merges two existing runs. Verified across all 11 realizable geometric update classes (75,000 updates and 5,400 state-position-input triples at $n = 6$ with zero discrepancies), the minimal forward-closed reader state is the coupled pair:

$$\text{Reader State} = (S, \Phi_S)$$

This cut algebra reveals how computational directionality (one-wayness) emerges directly from state compression (Theorem A″). While the forward transition $(S, \Phi_S) \xrightarrow{(p,u)} (S \cup \{p\}, \Phi_{S\cup\{p\}})$ is strictly functional for all cuts, the backward "un-reading" transition is non-functional for all $n \ge 3$. Specifically, consider the bottom-anchored cut $S = \{0,1\}$ and two assignments α and β that share $(a_0, b_0) = (0,0)$ at bit 0 (a carry-kill event) but differ at bit 1, with α having $(a_1, b_1) = (0,1)$ (propagate/transparent) and β having $(a_1, b_1) = (0,0)$ (kill). Because bit 0 is anchored at the bottom of the word and has no incoming carry, neither α nor β can ever emit a carry out of bit 1; thus, both compress to the identical bottom-anchored state:

$$\Phi_{\{0,1\}}(\alpha) = \Phi_{\{0,1\}}(\beta) = (\text{bot}, 0)$$

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However, if position $p = 0$ is subsequently "un-read" to reopen the interior cut $S' = \{1\}$, bit 1 is now exposed to a potential incoming carry from bit 0, causing the two histories to diverge into distinct equivalence classes:

$$\Phi_{\{1\}}(\alpha) = (\text{trans}) \neq \Phi_{\{1\}}(\beta) = (\text{int}, 1, 0)$$

In homological terms, a channel dies when its distinction becomes derivable from—or irrelevant to—the active equivalence vocabulary. Replacing a full microscopic history with a future-sufficient state discards interior propagate/kill distinctions that are shielded by a boundary anchor; when the inverse route attempts to reopen that boundary without global cofactor tracking, the discarded distinction becomes essential again, creating backward branching.

Schedule-Tail Bijection, Route-Space Reflection, and the Source-Fiber Selector Bound

To connect the local carry/borrow algebra and nested ring addresses across all 64 rounds of SHA-256, the global state space is parameterized through two exact macro-coordinate transformations: the rolling first-lane route word and the 16-word message-schedule tail.

The 2048-Bit Rolling Trace Reflection Law

In each round $t \in \{0, \dots, 63\}$ of SHA-256, the first working register updates to $U_t = T1_t + T2_t \pmod{2^{32}}$. Concatenating all 64 first-lane words forms the 2048-bit rolling route trace:

$$X = U_0 \| U_1 \| \cdots \| U_{63} \in \{0,1\}^{2048}$$

Because the SHA-256 state shift register pushes U_t down through registers a, b, c, d and into e, f, g, h over the final eight rounds ($t = 56 \dots 63$), subtracting the initial chaining state H_0 from the published digest D via Davies–Meyer feed-forward inversion ($Z = D - H_0 \pmod{2^{32}}$) uniquely and bijectively determines the final eight rolling lane sums $s(H) = (U_{56}, \dots, U_{63})$. Validated across six forward hashes and 10,000 arbitrary terminal states, this bijective 256-bit coordinate transformation maps the digest directly onto the 256-bit suffix of the route word X. Applying the width-2048 ring partner reflection P_{2048} to any non-palindromic route trace X immediately fixes its 256-bit prefix:

$$\text{prefix}_{256}(P_{2048}(X)) = \text{rev}_{256}(s(H))$$

This identity provides a direct, non-enumerative map from the published digest to a 256-bit address prefix on the reflected route in R_{2048}. However, empirical audits across seven known message traces and 100 arbitrary traces demonstrate why route reflection alone does not close the inversion: reflecting the 2048-bit trace $X \mapsto P_{2048}(X)$ destroys the internal SHA-256 message-schedule recurrence, failing all 48 schedule consistency equations ($W_{16} \dots W_{63}$). Similarly, applying the exact Möbius/ANF transform across the full 2-byte message domain (2^{16} inputs) for all 256 digest bits reveals that re-indexing the input space by ring address R_{16} does not lower algebraic degree: every output bit retains degree 15 or 16 with approximately 50% of all possible ANF monomials active. Thus, valid preimages are selected not by unconstrained ring reflection, but by the exact intersection of the ring-address boundary with the message-schedule recurrence.

Exact Coordinate Bijection of the Schedule Tail

While the 2048-bit route trace is over-parameterized relative to the 512-bit input block, the message schedule itself provides an exact 512-bit coordinate bijection. For rounds $16 \leq t \leq 63$, the forward SHA-256 schedule recurrence is defined by:

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$$W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \pmod{2^{32}}$$

Because W_{t-16} appears linearly with unit coefficient, solving for W_{t-16} yields the exact deterministic backward schedule recurrence:

$$W_{t-16} = W_t - \sigma_1(W_{t-2}) - W_{t-7} - \sigma_0(W_{t-15}) \pmod{2^{32}}$$

Iterating backward from $t = 63$ down to $t = 16$ proves that the 16-word schedule tail $T = (W_{48}, \dots, W_{63}) \in \{0,1\}^{512}$ uniquely reconstructs the entire schedule $W(T)$ and the initial 16-word message block $B = (W_0, \dots, W_{15}) \in \{0,1\}^{512}$. Verified across $1{,}000/1{,}000$ random schedule round-trips (reverse64_verified.py), the forward tail map $E: B \mapsto T$ is a strict mathematical bijection on $\{0,1\}^{512}$. Composing E^{-1} with the width-512 ring rank/unrank bijection R_{512} yields a lossless three-way coordinate equivalence:

$$T = (W_{48}, \dots, W_{63}) \;\leftrightarrow\; B = E^{-1}(T) \;\leftrightarrow\; i = \text{location}_{512}(E^{-1}(T))$$

In tail coordinates, the target digest fiber $\mathcal{F}_D$ for a single-block message class is defined by the exact conjunction of backward round closure and padding validity:

$$\mathcal{F}_D = \{T \in \{0,1\}^{512} \mid \text{Reverse}_{W(T)}(Z) = H_0 \;\wedge\; \text{PadValid}(E^{-1}(T))\}$$

The Pigeonhole Lower Bound on Source-Fiber Selection

Formulating the inverse problem in schedule-tail coordinates exposes a fundamental information-theoretic distinction between two different inversion objectives: **recovering the actual historical source** versus **computing any valid preimage**.

Consider the declared source class $\mathcal{X}$ of all single-block messages of exactly 55 bytes, which represents the maximum payload that fits within a single 512-bit SHA-256 block after appending the mandatory 0x80 padding byte and 64-bit big-endian length field ($55 \times 8 + 8 + 64 = 512$ bits). The domain cardinality is $|\mathcal{X}| = 256^{55} = 2^{440}$, whereas the SHA-256 digest space has cardinality 2^{256}. By the pigeonhole principle, restricting SHA-256 to $F: \mathcal{X} \to \{0,1\}^{256}$ forces the maximum fiber cardinality to satisfy:

$$\max_{D \in \{0,1\}^{256}} |F^{-1}(D)| \geq \left\lceil \frac{2^{440}}{2^{256}} \right\rceil = 2^{184}$$

Consequently, any deterministic reconstruction map $O_{\text{root}}(D, r(D)) = D_{\text{source}}$ guaranteed to identify the *actual historical member* of every digest fiber in the 55-byte class requires a retained source-selector coordinate $r(D)$ of width at least **184 bits** in the worst case ($\lceil \log_2 \max_D |F^{-1}(D)| \rceil \geq 184$). While the ring ordering R_{512} induces a canonical rank across the members of $\mathcal{F}_D$, the digest D alone cannot specify which rank originally occurred whenever $|\mathcal{F}_D| > 1$. Conversely, for short-interface domains where the message width is smaller than the digest width (such as the 8-bit Q, 16-bit GO, or 24-bit three-byte closure domains), the fiber generically collapses to a singleton ($|\mathcal{F}_D| = 1$), allowing pure constraint elimination via ROBDD cofactor reduction to recover the unique source without an external selector.

Comprehensive Program Ledger and Conclusions

Synthesizing all executed modules, algebraic proofs, and empirical audits establishes a rigorous division between completed structural theorems and active frontier links.

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System Component / Mathematical Claim	Verification Status	Evidence Basis & Scope
Symbolic Carry/Borrow Inverse Graph ($\Psi_H(\mu) = 0$)	**BOUND / RUNS**	$117{,}240$ (1B) and $118{,}294$ (2B) unique XOR/AND nodes; 256-zero replay check on Q and GO; $1{,}000/1{,}000$ 32-bit arithmetic checks
ROBDD Shannon Cofactor Preimage Closure (1-Byte)	**BOUND / RUNS**	Collapses $256 \to 1$ assignment in 7 return-bit constraints; recovers 0x51 (Q) at $R_8[70]$ across 4 variable orders
Ring Reflection Law ($R_L \circ \sigma_L = P_L \circ R_L$) & Rank/Unrank	**STANDS / RUNS**	Exhaustively verified for $L = 1 \ldots 12$; half-ring matches grow(L) for $L = 1 \ldots 10$; 512-bit rank/unrank round-trip verified
Ancestry Forest ($x = P^{\varepsilon}(F^k(\rho))$) & Generating Series	**STANDS / RUNS**	Reconstructs all $8{,}190$ words ($L = 1 \ldots 12$) and 512-bit roots; exact rational generating functions for $\lvert z \rvert < 1/2$ [cite: 1, 4]
DI Chain Recurrences & 5-Bit Rollover (0x1BB vs D1B)	**BOUND / RUNS**	$56/56$ audit checks passed; $G_{L,i} = B_{L,(i+3) \bmod 8}$; mod-255 doubling orbit at $L = 8$; exact 10-bit packing resolved
Möbius / ANF Transducer Bridge ($MR = TM, MP = \tilde{P}M$)	**STANDS / RUNS**	Proved via Pascal's identity mod 2 ($T = I + S, UTU = T^{-1}$); $256/256$ bytes and $32{,}768/32{,}768$ window reads verified
Homological Cut Normal Form ($\dim M_{ij} = n - j$) & Theorem A″	**STANDS / RUNS**	All-n proof via L1 low-bit transport and L2 staircase decomposition; $75{,}000$ updates at $n = 6$; backward non-invertibility proved
Schedule-Tail Bijection ($E: B \leftrightarrow T$) & 184-Bit Selector Bound	**STANDS / RUNS**	Exact theorem from backward schedule recurrence ($1{,}000/1{,}000$ round-trips); pigeonhole bound $2^{440}/2^{256} = 2^{184}$ for 55-byte class

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System Component / Mathematical Claim	Verification Status	Evidence Basis & Scope
Radial Reader Pre-p Profile ($q_D = 100\,1^{D-5}00$)	**STANDS (Pre-p) / OPEN (Tail)**	Pre-p center-word bit locations algebraically derived; generated-word identities for $p \le L \le D$ and uniform $D \equiv 2 \pmod 3$ remain **OPEN** [cite: 1, 3]
Region-Level ROBDD / Tail Pruning for Wide SHA-256 Interfaces	**OPEN**	Extending ROBDD cofactor elimination with schedule-tail region certificates to prune multi-byte address prefixes before full expansion

Conclusions and Next Structural Directives

The symbolic DAG experiments and ROBDD execution records definitively prove that SHA-256 inversion cannot be closed by local gate propagation (0 bits bound), because correlation constraints across carry chains and schedule expansions are stored in reconvergent nonlinear loops. Global Shannon cofactor reduction replaces local unit propagation as the exact binding mechanism: canonical ROBDD cofactor conjunction resolves reconvergent carry and borrow paths simultaneously, collapsing the 1-byte preimage space to its unique solution (0x51 at $R_8[70]$) using only the first seven bits of the returned initial vector.

Simultaneously, coordinate commutation unifies the multiscale readout stack. The nested binary ring R_L, the rolling DI window stack, and the Boolean Möbius difference engine are formally linked by equivariant operators ($R_L \circ \sigma_L = P_L \circ R_L$, $MR = TM$, and $MP = \tilde{P}M$). Homological cut algebra further establishes that any compressed reader must carry its cut geometry alongside its quotient state (S, Φ_S), as discarding boundary-shielded interior distinctions is the exact mechanism that generates backward branching.

Finally, because the 512-bit schedule tail $T = (W_{48}, \ldots, W_{63})$ is bijective with the initial message block B and losslessly addressable via R_{512}, the open frontier for multi-byte and full-block preimage routing is sharply bounded. Future computational execution must embed the schedule-tail recurrence and PadValid constraints directly into the ROBDD cofactor schedule, deriving region-level address exclusion certificates that prune incompatible subtrees of R_{512} before intermediate decision-node counts scale exponentially.

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